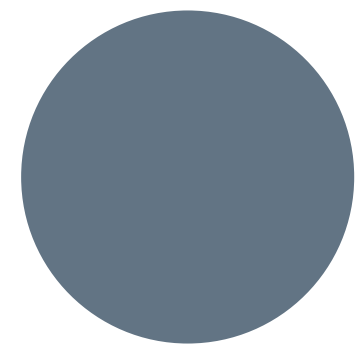
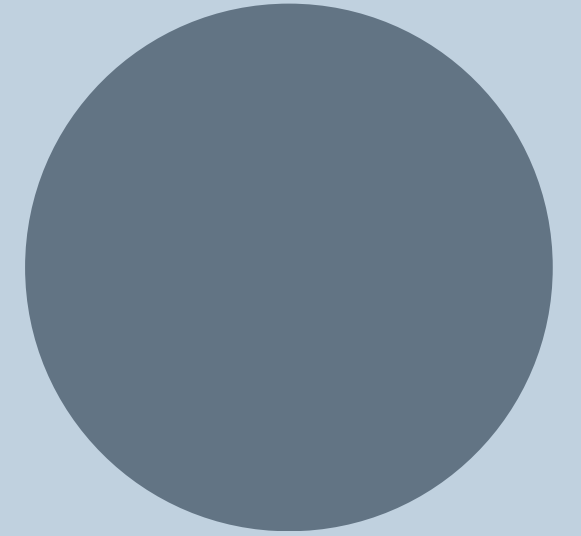

TRAFFIC AND ACCIDENT DETECTION SYSTEM



25-26J-174



MEET OUR TEAM



Pushpamal K P N



Wathsalavi K HS

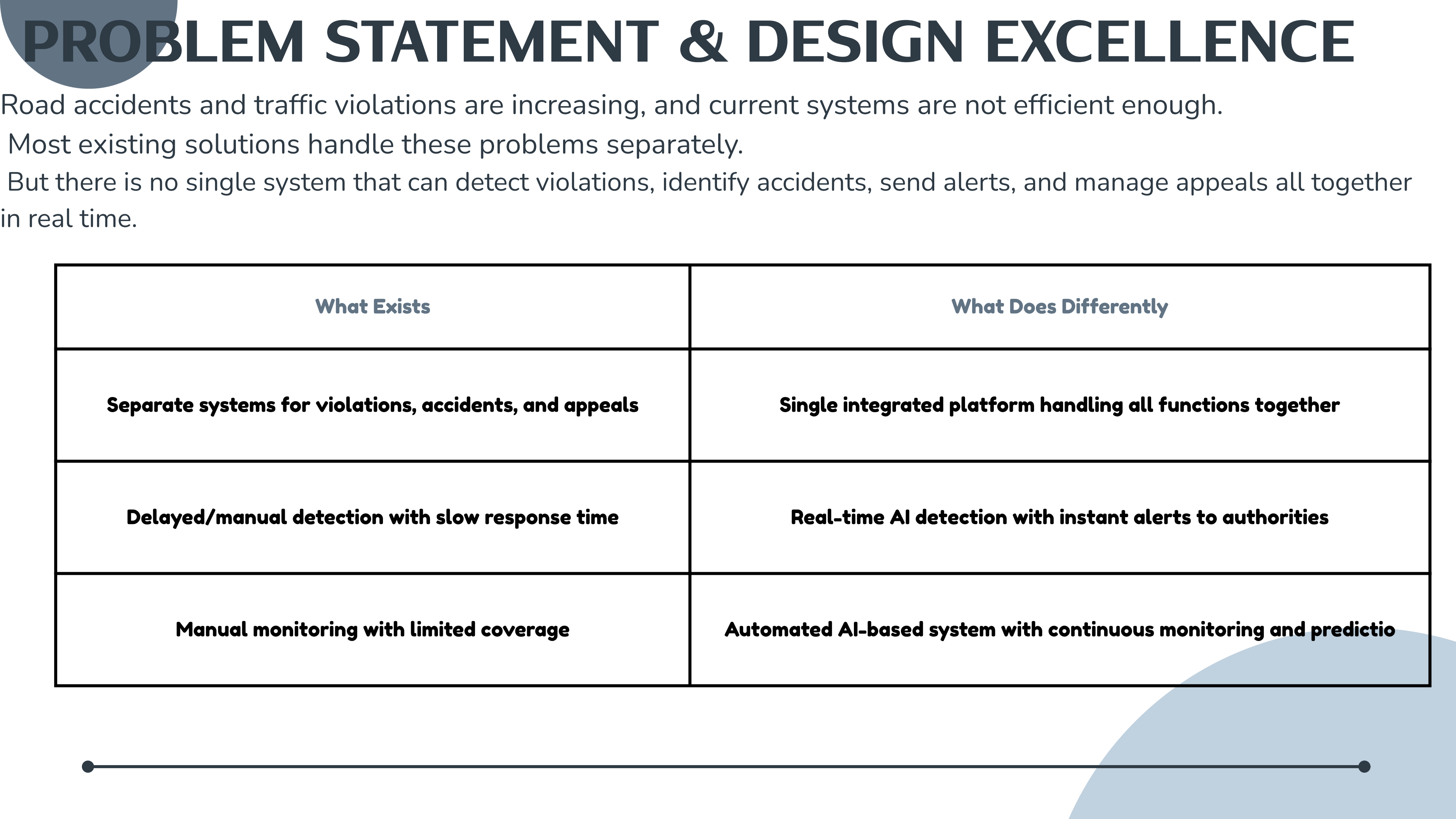


Shehara I.G.D.S



Rashmi B.G.R





PROBLEM STATEMENT & DESIGN EXCELLENCE

Road accidents and traffic violations are increasing, and current systems are not efficient enough.
Most existing solutions handle these problems separately.
But there is no single system that can detect violations, identify accidents, send alerts, and manage appeals all together in real time.

What Exists	What Does Differently
Separate systems for violations, accidents, and appeals	Single integrated platform handling all functions together
Delayed/manual detection with slow response time	Real-time AI detection with instant alerts to authorities
Manual monitoring with limited coverage	Automated AI-based system with continuous monitoring and predictio

THE ACTIONS TAKEN FOR A PP2 COMMENTS

Comment

- * inadequate demo and unfamiliar with most aspects of the code
- * unable to answer basic questions from the panel
- * core objective of the program seems to work, however, the mechanisms and the reasoning behind why certain features in the application are used are not known

Action taken

Improved system knowledge, practiced the demo, and strengthened understanding of concepts to confidently explain features and answer panel questions.

Comment

- * inadequate knowledge of the core principles being used in the program
- * the developed predication for the appeal uses synthetic data (which is okay) but a concrete plan for how to obtain real world data is not available

Action taken

Improved understanding of the core principles used in the AI-based appeal prediction system through additional research and technical discussions.

Prepared a concrete plan to collect real-world traffic violation and appeal data through public datasets, authority collaboration, and future system-generated records for model improvement.

THE ACTIONS TAKEN FOR A PP2 COMMENTS

Comment

- * student shows inadequate knowledge in the underlying function of the program
- * the novelty of the project could be significantly higher

Action taken

expands the scope from 1 to 2 violations

Comment

- * the project does not appear to have any novel aspects in a way that fulfills the research requirements
- * certain minimal features have been added, however they lack the scope and depth of a 1 year research project

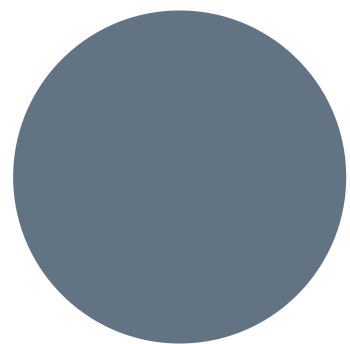
Action taken

Improved the helmet violation detection and number plate identification component by integrating the trained helmet detection model with the overall system. When a helmet or other traffic violation is detected, the system captures evidence and reads the vehicle number plate.

Traffic Violation Detection System



Pushpamal K P N
IT22561916



TRAFFIC LIGHT VIOLATION & SEATBELT DETECTION

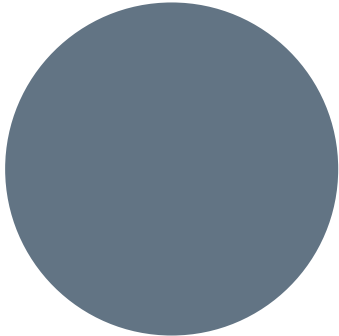
Key Functions

- Processes live or uploaded video feeds from intersection cameras.
- Detects traffic light states (red, yellow, green) and stop-line positions using YOLOv8m.
- Flags a red-light violation when a vehicle crosses the stop-line after the signal turns red.
- Detects driver and front passenger using lightweight YOLOv8n model.
- Classifies seatbelt status: worn / not worn for both driver and passenger.
- Captures vehicle evidence images.

Contribution

- **Reducing manual inspection** – Automatically identifies red-light runners and unbelted occupants without human intervention.
- **Improving road safety** – Deters red-light violations and encourages seatbelt use through reliable, real-time detection.
- **Enabling accurate evidence collection** – Saves timestamped, annotated images for citation and appeal processes.
- **Supporting scalability** – Runs concurrently on standard GPU hardware, suitable for city-wide deployment.

Helmet Violation & Number Plate Detection



HELMET VIOLATION & NUMBER PLATE DETECTION

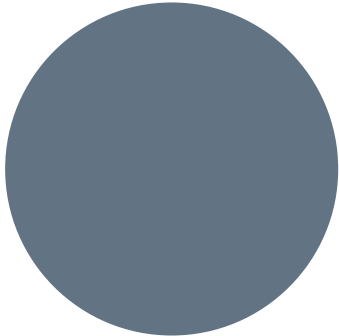
Key Functions

- Detects motorcycle riders in surveillance footage.
- Identifies helmet and non-helmet cases.
- Flags non-helmet riders as violations.
- Captures vehicle evidence images.
- Extracts license plate details.
- Sends violation data for enforcement.

Contribution

This subsystem supports automated traffic law enforcement by improving rider safety monitoring, reducing manual inspection, and enabling accurate identification of helmet-related violations.

Citizen Appeal & Digital Transparency System



CITIZEN APPEAL & DIGITAL TRANSPARENCY SYSTEM

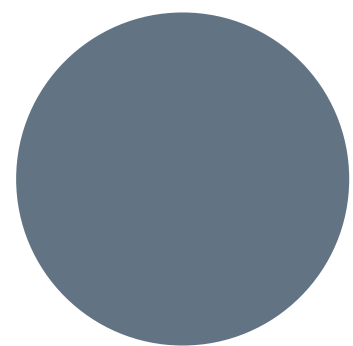
Key Functions

- Allows citizens to submit and track traffic violation appeals digitally.
- Provides real-time appeal status updates for transparency.
- Uses AI prediction models to analyze appeal outcomes and patterns.
- Detects high-priority or suspicious appeal cases automatically.
- Reduces manual paperwork and improves processing efficiency.
- Maintains secure digital records of appeal history and evidence.
- Generates analytical insights for traffic law enforcement authorities.

Contribution

This subsystem enhances transparency and accountability in traffic violation management by enabling citizens to monitor appeal progress digitally while supporting authorities with AI-driven prediction and decision-support mechanisms. It reduces delays, minimizes manual intervention, and improves the overall efficiency and fairness of the appeal handling process.

Accident Severity Assessment & Emergency Alert System



ACCIDENT SEVERITY ASSESSMENT & EMERGENCY ALERT SYSTEM

Key Functions

- Detects accidents using collision, speed drop & tracking
- Estimates vehicle speed before impact
- Classifies severity: Minor / Moderate / Severe
- Detects fire & smoke at scene
- Sends real-time alerts to hospitals, police & ambulance
- Includes location, time & severity details

Contribution

- Developed accident detection logic using tracking & motion analysis
- Built severity classification model
- Integrated fire detection module
- Implemented real-time alert system
- Connected system with emergency services
- Improved response time for accidents

RESULTS & TESTIMONIALS

Performance

- High accuracy in detection models (~90%+)
- Strong F1 Score for classification tasks
- Real-time processing with fast response

User Feedback

- Easy to use and reliable system
- Faster detection and alerting
- Improved transparency in appeals



LIMITATIONS & FUTURE IMPROVEMENTS

Limitations:

- Performance depends on camera quality
- Night-time detection accuracy can reduce
- Heavy traffic may affect tracking accuracy

Future Improvements:

- Improve night vision detection
- Add GPS-based accident auto-location
- Mobile app integration for citizens
- Cloud-based real-time dashboard





**SRI LANKA
INSTITUTE OF INFORMATION TECHNOLOGY**

16th Floor, BoC Merchant Tower, No. 28, St. Michael's Road, Colombo 03

Date: 08/09/2025

Your Ref :

My Ref : 25-26J-174

Mr. K. G. P. Priyadarshana,
Officer in Charge (OIC),
Baddegama Police Station 559J+89W,
Hikkaduwa Road,
Baddegama 80200.

Dear Sir,

Request for Assistance in Data Collection for Final Year Research Project

The Sri Lanka Institute of Information Technology (SLIIT) is the largest Degree Awarding Institute in the field of information Technology recognized by the University Grants Commission under the Universities Act. It was established in the year 1999 to educate and train Information Technology (IT) Professionals required by the fast-growing IT Industry in Sri Lanka.

We would like to introduce the following group of final year undergraduates from the Sri Lanka Institute of Information Technology (SLIIT), who are currently conducting research titled "Traffic and Accident Detection System" as part of their B.Sc. in Information Technology Degree program. Under the guidance of Mr. Dinith Primal, they are dedicated to exploring innovative solutions in this domain:

- IT22052360 – Rashmi B.G.R
- IT22338334 - Shehara I.G.D.S
- IT22561916 - Pushpamal K.P.N
- IT22305596 - Wathsalavi K.H.S

We kindly request your assistance in facilitating their data collection process within your organization to enhance their research dataset. Should you have any inquiries or require further clarification regarding the project, please feel free to reach out to us.

Thank you for your cooperation.

Dr. Jayantha Amararachchi
Assistant Professor/
Research Project Coordinator,
jayantha.a@slit.lk
+94 11 754 4103

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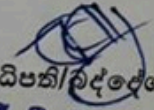
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ශීර්ෂගත කරුණ හා සබැඳි ඔබගේ ඉල්ලීම පරිදි පහත නම් සඳහන් ශිෂ්‍ය හිෂ්‍යාවන් සිව්දෙනා බද්දේගම පොලිස් ස්ථානය වෙත පැමිණ රිය අනතුරු විමර්ශනය සම්බන්ධයෙන් සජීවීව සම්බන්ධ විමට අවසර ඉල්ලා සිටින ලදී. කරුණු එසේමුත් දැනට පවතින නෛතික ප්‍රතිපාදන මත සජීවීව සිවිල් පුරවැසියෙකුට රිය අනතුරක් විමර්ශනය සඳහා දායකත්වය ලබා දීමට නොහැකි අතර ප්‍රායෝගිකව රිය අනතුරු විමර්ශනය සම්බන්ධ දැනුම න්‍යායාත්මක දැනුම ලබා දීමට කටයුතු කලෙමි.

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02	අයි.පී.ඩී.එස් ශෙහාරා	IT22338334
03	කේ.පී.එන් පුෂ්පමාලි	IT22561916
04	කේ.එච්.එස් චන්සලාචි	IT22305596


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**Thank
You**